

Rider Behavior: Q1 2019 vs Q1 2020

Understanding how members and casual riders use bike-share differently to support membership conversion strategy.



TOTAL RIDES

792K

791,955 records

MEMBER RIDES

720K

91.0% of rides

CASUAL RIDES

72K

9.0% of rides

AVG RIDE LENGTH

18.0

minutes

Executive summary

Divvy public trip data, Q1 2019 and Q1 2020, combined into one dataset of 791,955 rides.

Members drive volume; casual riders show longer, more leisure-oriented behavior

Business question: How do annual members and casual riders use bikes differently?

TOTAL RIDES

791,955

Final analysis file

MEMBER RIDES

720K

91.0% of rides

CASUAL RIDES

72K

9.0% of rides

AVERAGE LENGTH

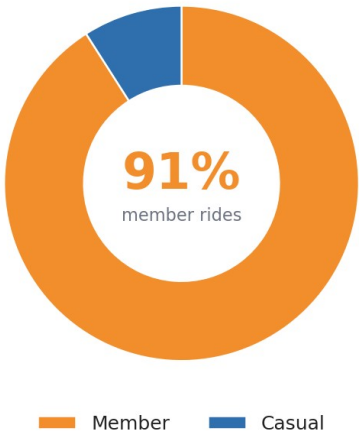
18.0 min

Median: 9.0 min

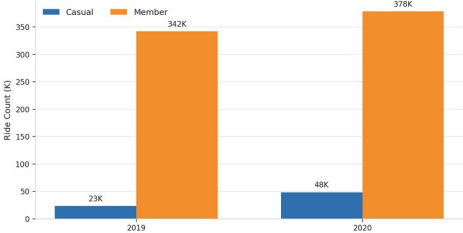
Executive answer

- Members dominate ride volume: 91.0% of all rides are from members.
- Casual riders take longer rides: their median ride is about 2.5 times a member's (21–23 vs 8–9 minutes).
- Casual usage is weekend-skewed: Sunday and Saturday carry the highest casual ride share.
- Members show routine hourly peaks: strongest activity occurs around 08:00 and 17:00.

Rider Mix



Members Generate Most Rides



Business task and decision context

Cyclistic wants to convert casual riders into annual members using data-backed marketing strategy.

Primary question assigned

How do annual members and casual riders use Cyclistic bikes differently?

This deck focuses on behavior differences that can inform membership conversion tactics.

Stakeholders

- **Lily Moreno:** Director of Marketing; owns campaign strategy.
- **Marketing analytics team:** prepares analysis and reporting.
- **Executive team:** approves recommendations and investment.

Required case study deliverables

1 Business task

2 Data sources

3 Cleaning documentation

4 Analysis summary

5 Visual findings

6 Top 3 recommendations

Source: Case study brief assigns the first business question and requires six deliverables.

Prepare

Define the data, scope, and toolchain used in the analysis.

Data source, scope, and analysis file

The Q1 2019 and Q1 2020 trip files were combined into one dataset and visualized in Tableau and Power BI.

2019 RECORDS

365,069

Divvy Trips Q1 2019

2020 RECORDS

426,886

Divvy Trips Q1 2020

COMBINED RECORDS

791,955

Appended dataset

RIDER TYPES

2

member and casual

Fields used for the story

- ride_id, started_at, ended_at
- member_casual, ride_length_minute
- day_of_week, day_name, start_hour
- year and station names for context

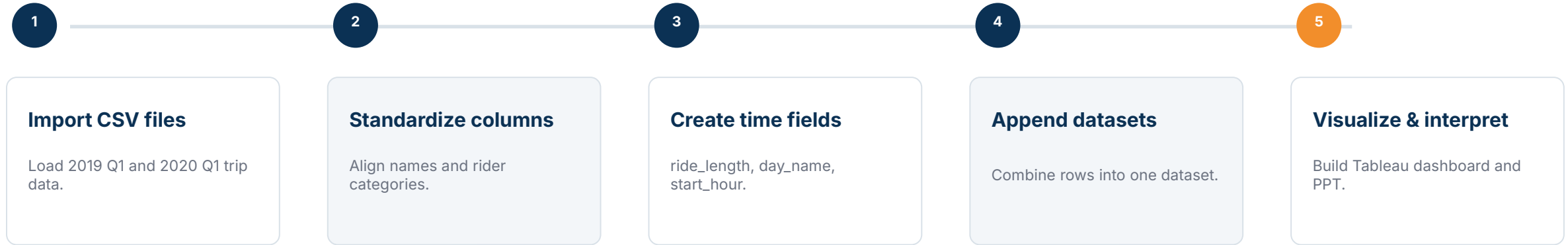
Privacy note: no personally identifiable rider data is used.

Scope boundaries

- This is a Q1 2019 vs Q1 2020 comparison, not a full-year seasonality analysis.
- Ride length was calculated in Google Sheets before the files were combined in Python.
- No filter for negative durations or Divvy test rides is applied, so averages include outliers; medians are used for comparisons.

Analysis workflow

Python/Colab was used to prepare the combined dataset; Tableau was used to create the dashboard.



Documentation statement

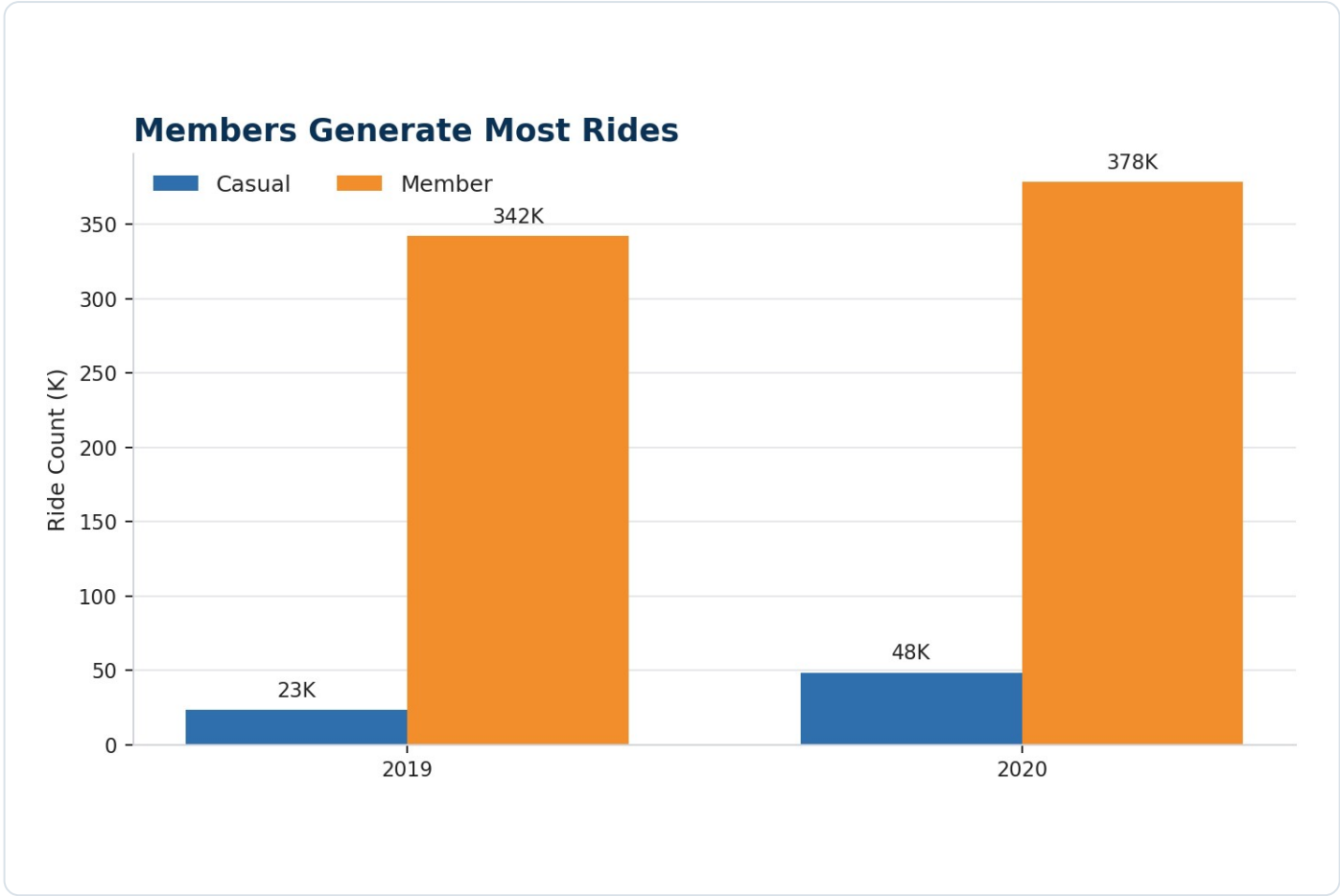
The final dataset contains 791,955 rows and 15 fields. It supports analysis of ride volume, ride duration, weekday distribution, and hourly usage patterns by rider type and year.

Finding 1

Ride volume by rider type and year

Members generate the overwhelming majority of rides

In the final dataset, members account for 720K rides, while casual riders account for 72K rides.



What the chart shows

- Member rides: 720,313 (91.0%).
- Casual rides: 71,642 (9.0%).
- Members make roughly 10× more trips than casual riders.
- This confirms that member behavior is more frequent and routine by volume.

Implication

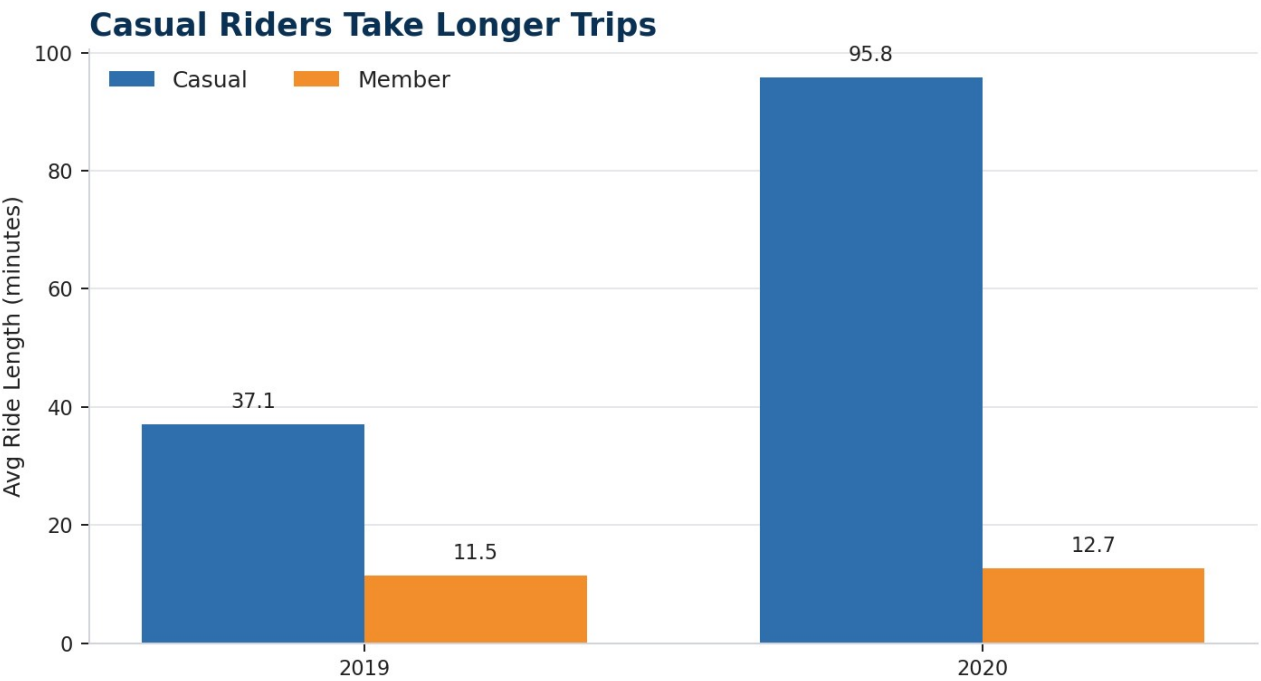
Growth does not require explaining the service from scratch: casual riders already use the system and can be targeted for conversion.

Finding 2

Average and median ride length by rider type

Casual riders take longer trips than members

Casual riders ride longer in both years. Medians are the safer comparison because a few very long casual trips pull the averages up.



CASUAL AVG

76.8 min

Median 22.0 min

MEMBER AVG

12.1 min

Median 8.5 min

Values by year

- 2019 casual avg: 37.07 min; member avg: 11.49 min.
- 2020 casual avg: 95.78 min; member avg: 12.68 min.
- This supports messaging that targets casual riders with long-trip value propositions.

Note: the 2020 casual average is inflated by outliers (casual rides starting at 4 AM average over 20 hours). Medians: casual 23.4 / 21.2 min, member 8.4 / 8.6 min (2019 / 2020).

Finding 3

Weekday distribution by rider type

Casual riding is more concentrated on weekends

Casual riders over-index on Sunday and Saturday; members concentrate more on weekdays.



Weekend skew

CASUAL SUNDAY	CASUAL SATURDAY
26.0%	18.8%
highest share	second-highest
MEMBER TUESDAY	MEMBER WEEKEND
17.8%	16.6%
weekday peak	Sun + Sat

Implication

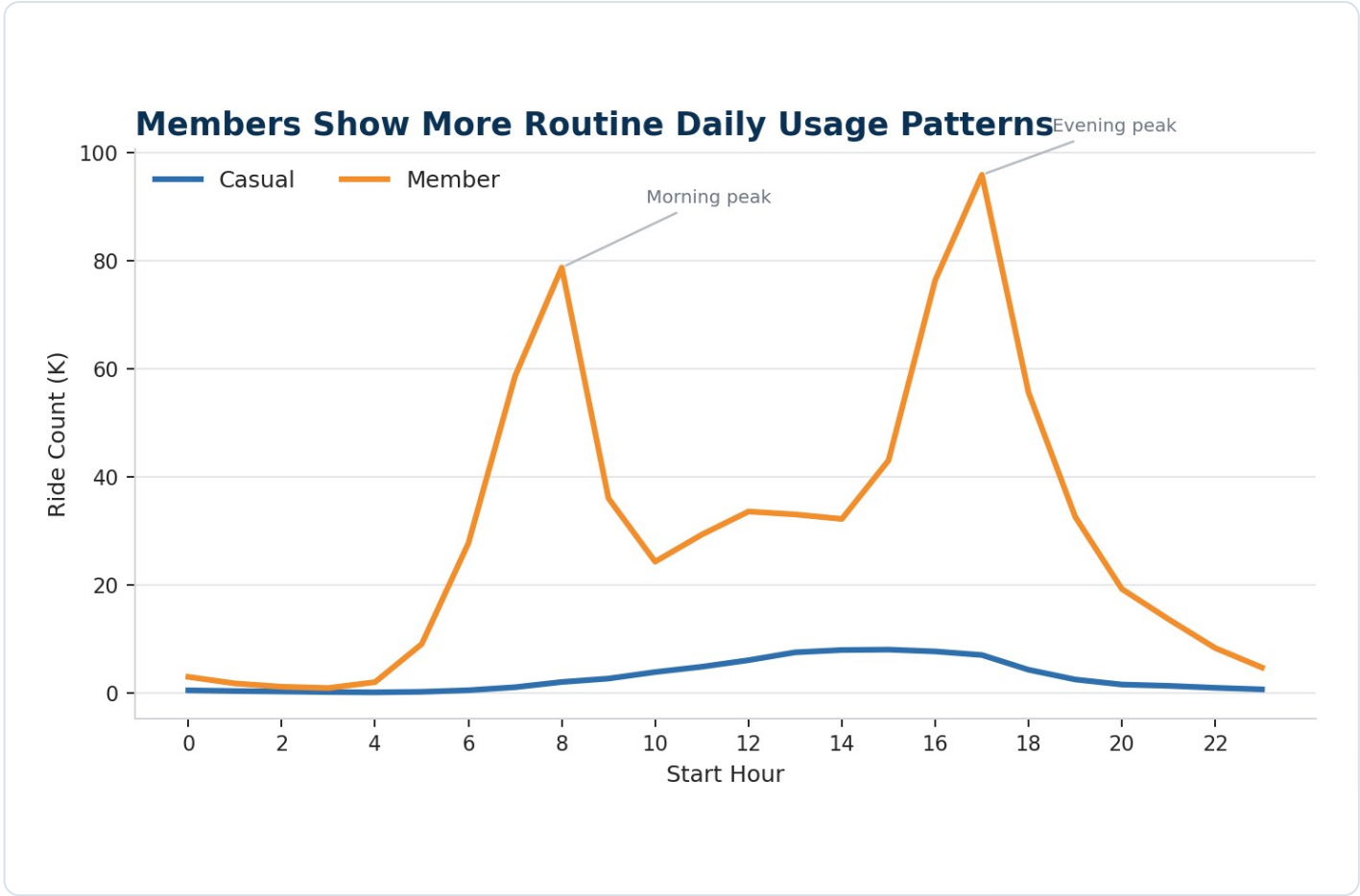
Weekend touchpoints are more relevant for casual conversion than broad, all-week messaging.

Finding 4

Hourly usage patterns by rider type

Members show stronger routine hourly usage patterns

Member activity has clear morning and evening peaks; casual activity peaks later in the day.



Peak start hour (both years combined)

MEMBER PEAK

17:00

95,865 rides

CASUAL PEAK

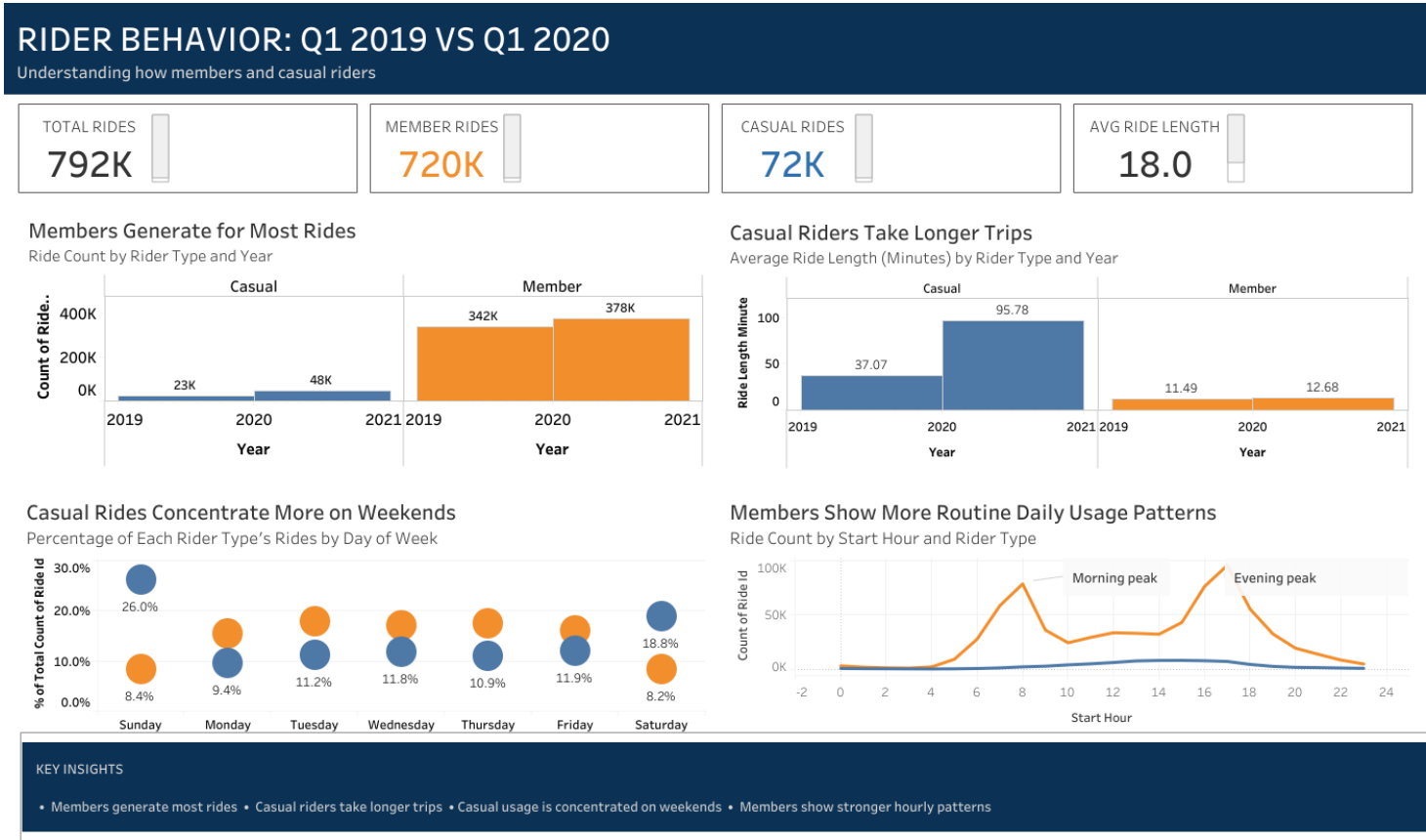
15:00

7,999 rides

- Members display sharper 08:00 and 17:00 peaks.
- Casual rides build through midday and afternoon.
- Use "commute-related" as an inference, not a confirmed cause.

Dashboard view: rider behavior comparison

The dashboard combines KPIs, volume, duration, weekday share, and hourly usage in one executive view.



Top three recommendations

The recommendations focus on converting casual riders who already show high engagement.

1 Weekend conversion campaign

Target casual riders on Saturday and Sunday with membership trial offers and post-ride prompts.

Why: casual ride share is highest on Sunday and Saturday.

Measure: CTR, conversion rate, cost per conversion, repeat rides.

2 Long-trip value messaging

Show savings or benefits of membership after longer casual rides and repeat long-duration sessions.

Why: casual rides are longer on average and median than member rides.

Measure: CTR, conversion rate, cost per conversion, repeat rides.

3 Segmented digital experiments

Run A/B tests by rider type, day, start hour, and high-casual stations; measure conversion rate.

Why: usage patterns are different enough to justify targeted campaigns.

Measure: CTR, conversion rate, cost per conversion, repeat rides.

Use the findings, but avoid over-claiming causality

The data answers usage differences, not the exact reason casual riders would buy a membership.

Limitations

- Scope is Q1 2019 vs Q1 2020, not a full-year view.
- No rider-level identity: repeat behavior cannot be directly tracked.
- No pricing, revenue, or membership conversion data is included.
- Trip purpose is inferred from patterns, not observed directly.
- Weather, events, and operational changes are not controlled.

Next steps

- Extend analysis to a full 12-month dataset for seasonality.
- Add weather and station-level factors to explain demand variation.
- Create segments for repeat casual riders, if privacy-safe IDs exist.
- Run controlled digital campaign tests against the identified segments.
- Connect campaign exposure to membership conversion outcomes.

Final conclusion: Members use the service more frequently and routinely; casual riders use it less often but take longer and more weekend-oriented trips.

Metric reconciliation

These are the key values used in the deck and dashboard.

Metric	Value
Total rides	791,955
Member rides	720,313 (91.0%)
Casual rides	71,642 (9.0%)
Overall average ride length	18.0 minutes
Overall median ride length	9.0 minutes
2019: casual / member rides	23,163 / 341,906
2020: casual / member rides	48,479 / 378,407
2019: casual / member average duration	37.07 / 11.49 minutes
2020: casual / member average duration	95.78 / 12.68 minutes

Data source: Trip_Q1_2019_vs_Trip_Q1_2020.csv, built in notebooks/cyclistic_bike_share_analysis.ipynb. Aggregates are in data/summary/.